

Quantitative Business Modeling

Sensitivity Analysis

Sensitivity analysis is the study of how changes in the coefficients of a linear programming problem affect the optimal solution.

Using sensitivity analysis, we can answer two things:

1. How will a change in an *objective function coefficient* affect the optimal solution?
2. How will a change in a *right-hand-side value* for a constraint affect the optimal solution?

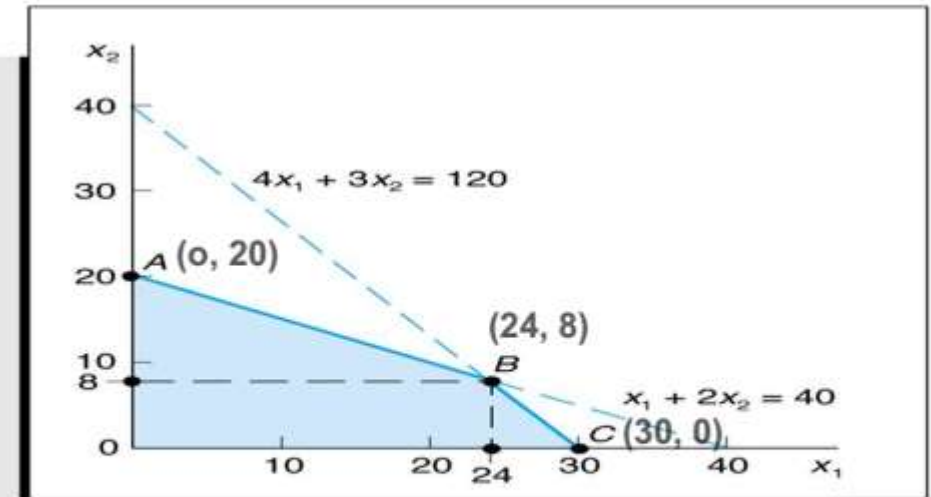
Example:

Maximize $Z = \$40x_1 + \$50x_2$

subject to: $1x_1 + 2x_2 \leq 40$

$4x_1 + 3x_2 \leq 120$

$x_1, x_2 \geq 0$



$$Z(A) = 40 \cdot 0 + 20 \cdot 50 = \$1000$$

$$Z(B) = 40 \cdot 24 + 50 \cdot 8 = \$1360$$

$$Z(C) = 40 \cdot 30 + 50 \cdot 0 = \$1200$$

Sensitivity Analysis

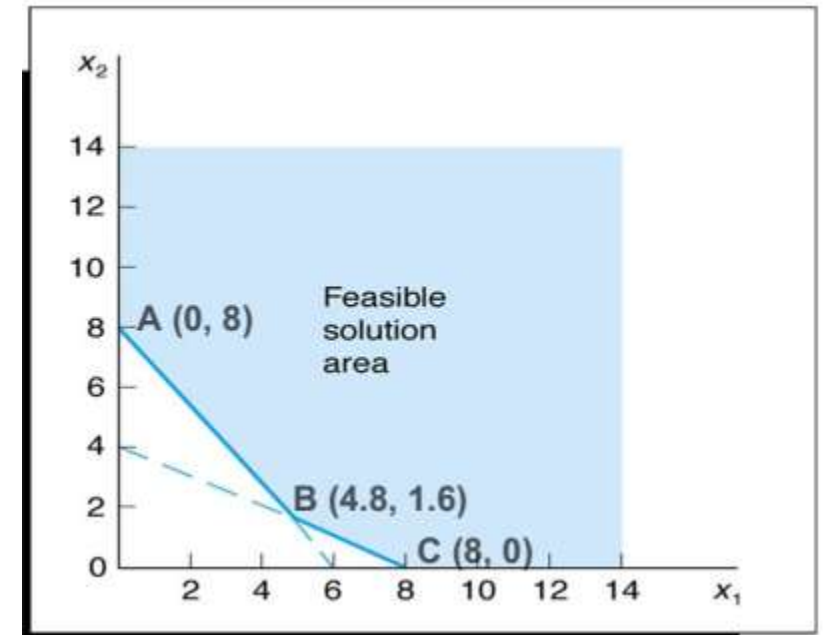
- ✓ Sensitivity analysis only begin after the optimal solution to the original LPP has been obtained.
- ✓ Therefore, referred as “*Postoptimality Analysis*”.

Application of Sensitivity Analysis

- Real-world problems exist in a changing environment – [Raw materials price change, raw material availability change, demands change, production capacities change]
- If LPP used in such an environment, some of the coefficients in the model change over time.
- Sensitivity analysis determines how these changes affect the optimal solution.

Example:

$$\begin{aligned} \text{Minimize } Z &= \$6x_1 + \$3x_2 \\ \text{subject to: } & 2x_1 + 4x_2 \geq 16 \\ & 4x_1 + 3x_2 \geq 24 \\ & x_1, x_2 \geq 0 \end{aligned}$$



$$Z(A) = 6 \cdot 0 + 8 \cdot 3 = 24$$

$$Z(B) = 6 \cdot 4.8 + 3 \cdot 1.6 = 33.6$$

$$Z(C) = 6 \cdot 8 + 3 \cdot 0 = 48$$

Sensitivity Analysis

RMC, Inc., produces a variety of chemical-based products. In a particular production process, three raw materials are used to produce two products: a fuel additive and a solvent base. The fuel additive is sold to oil companies. The solvent base is sold to a variety of chemical firms. The three raw materials are blended to form the fuel additive and solvent base as indicated in below Table, which shows that a ton of fuel additive is a mixture of 0.4 tons of Material 1 and 0.6 tons of Material 3. A ton of solvent base is a mixture of 0.5 tons of Material 1, 0.2 tons of Material 2, and 0.3 tons of Material 3. profit contribution of \$40 for every ton of fuel additive produced and \$30 for every ton of solvent base produced. Let us now use linear programming to determine the number of tons of fuel additive and the number of tons of solvent base to produce in order to maximize total profit contribution.

MATERIAL REQUIREMENTS PER TON FOR THE RMC PROBLEM

	Product	
	Fuel Additive	Solvent Base
Material 1	0.4	0.5
Material 2		0.2
Material 3	0.6	0.3

Material	Amount Available for Production
Material 1	20 tons
Material 2	5 tons
Material 3	21 tons

Sensitivity Analysis

X = number of tons of fuel additive produced

Y = number of tons of solvent base produced

$$\text{Max } 40X + 30Y$$

$$0.4X + 0.5Y \leq 20$$

$$0.2Y \leq 5$$

$$0.6X + 0.3Y \leq 21$$

$$X, Y \geq 0$$

$$\text{Max } 40X + 30Y$$

$$4X + 5Y \leq 200$$

$$2Y \leq 50$$

$$6X + 3Y \leq 210$$

$$X, Y \geq 0$$

Sensitivity Analysis

1. How will a change in an *objective function coefficient* affect the optimal solution?

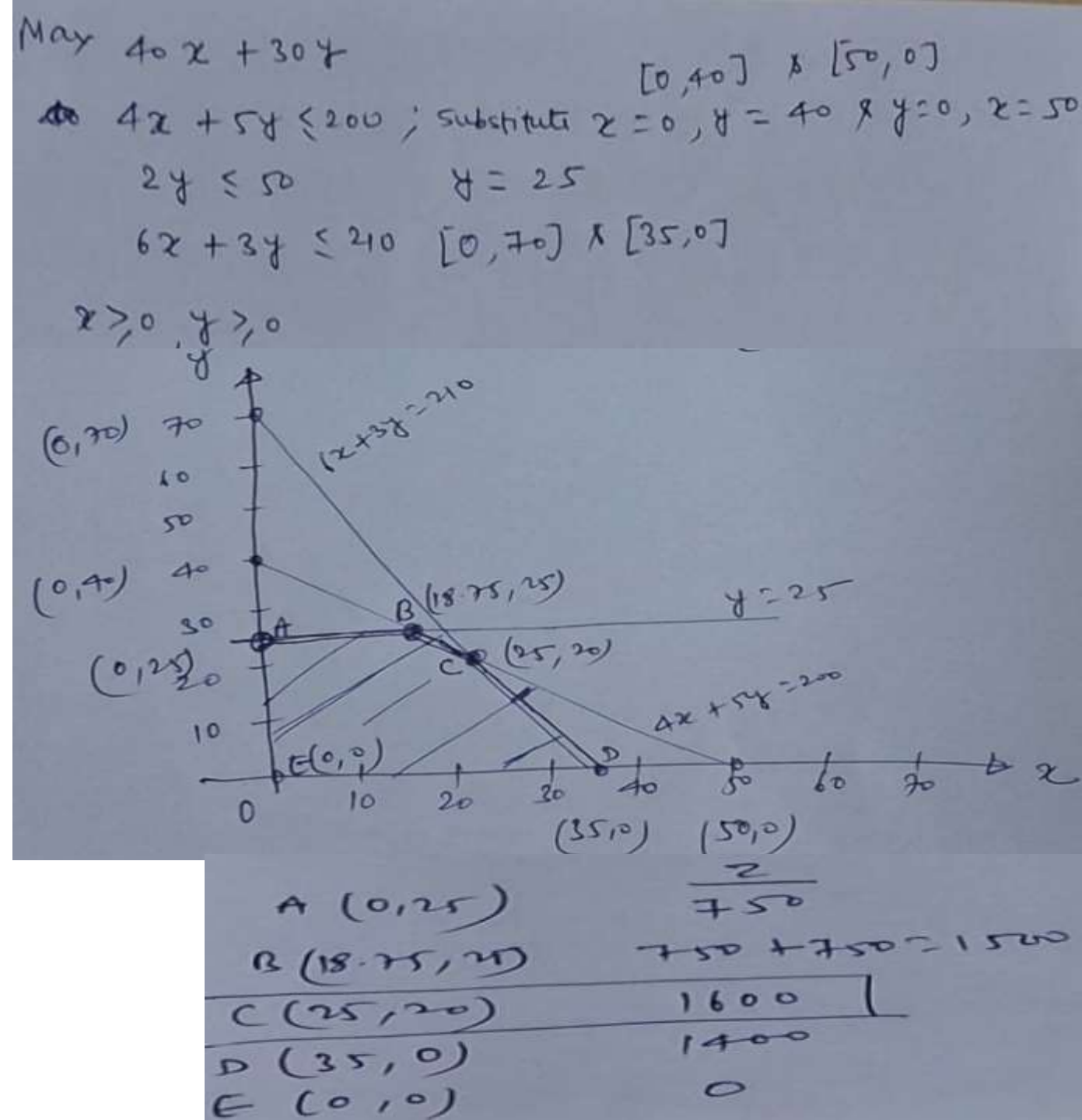
Optimal solution: $X = 25$ tons & $Y = 20$ tons, profit = \$1600.

Suppose profit contribution for the fuel additive fall from \$40 to \$30/ton.

Question is-

Whether producing 25 tons of fuel additive and 20 tons of solvent base is still best??

Answer by Sensitivity Analysis



Sensitivity Analysis

$$\text{Max } 30X + 30Y$$

$$4X + 5Y \leq 200$$

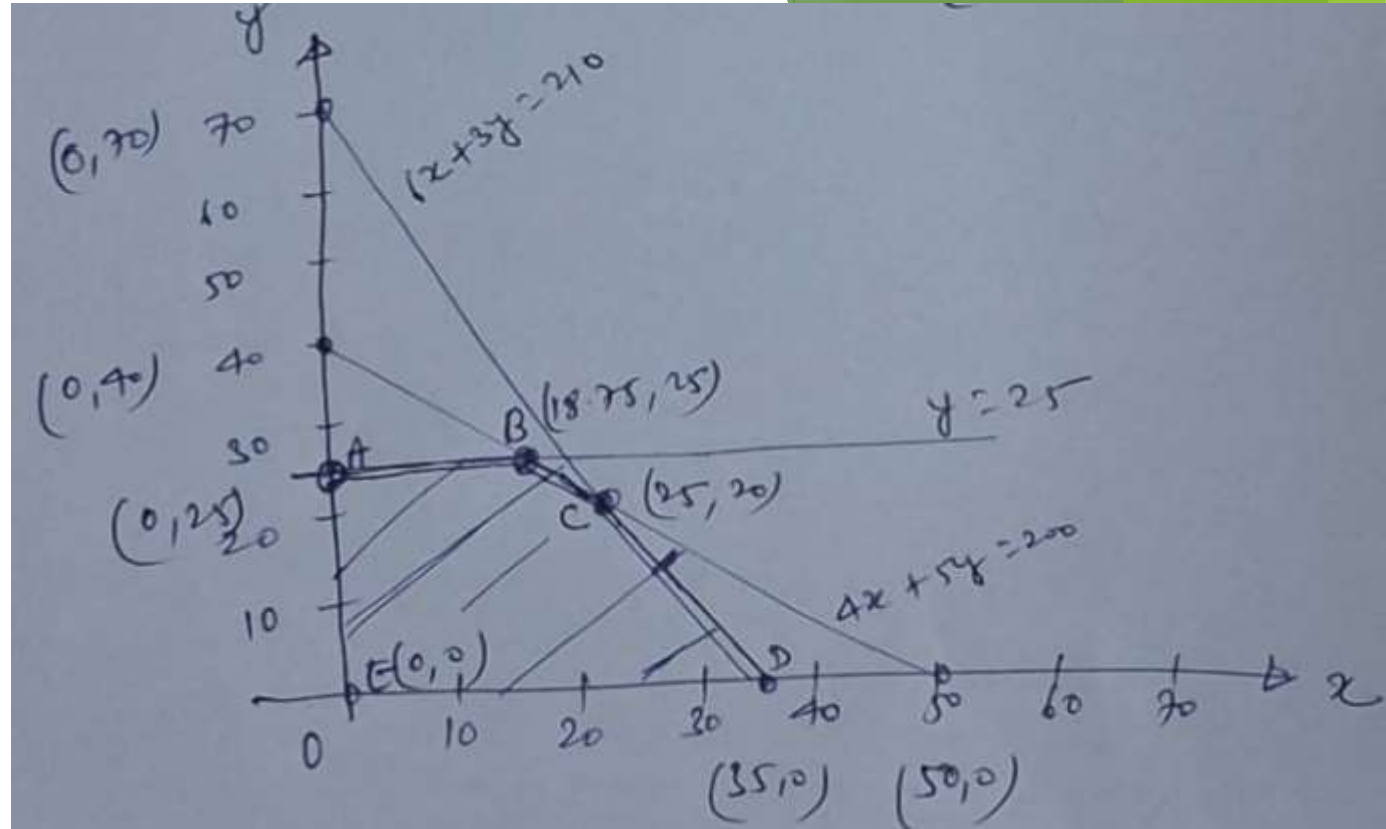
$$2Y \leq 50$$

$$6X + 3Y \leq 210$$

$$X, Y \geq 0$$

Feasible region for the revised problem remains the same. Because constraints are same

Decision variables value is same (Producing $X=25$ & $Y=20$ is still optimal), however objective value changed



Sensitivity Analysis

Suppose profit contribution for the fuel additive further reduced to \$20 per ton.

$$\text{Max } 20X + 30Y$$

$$4X + 5Y \leq 200$$

$$2Y \leq 50$$

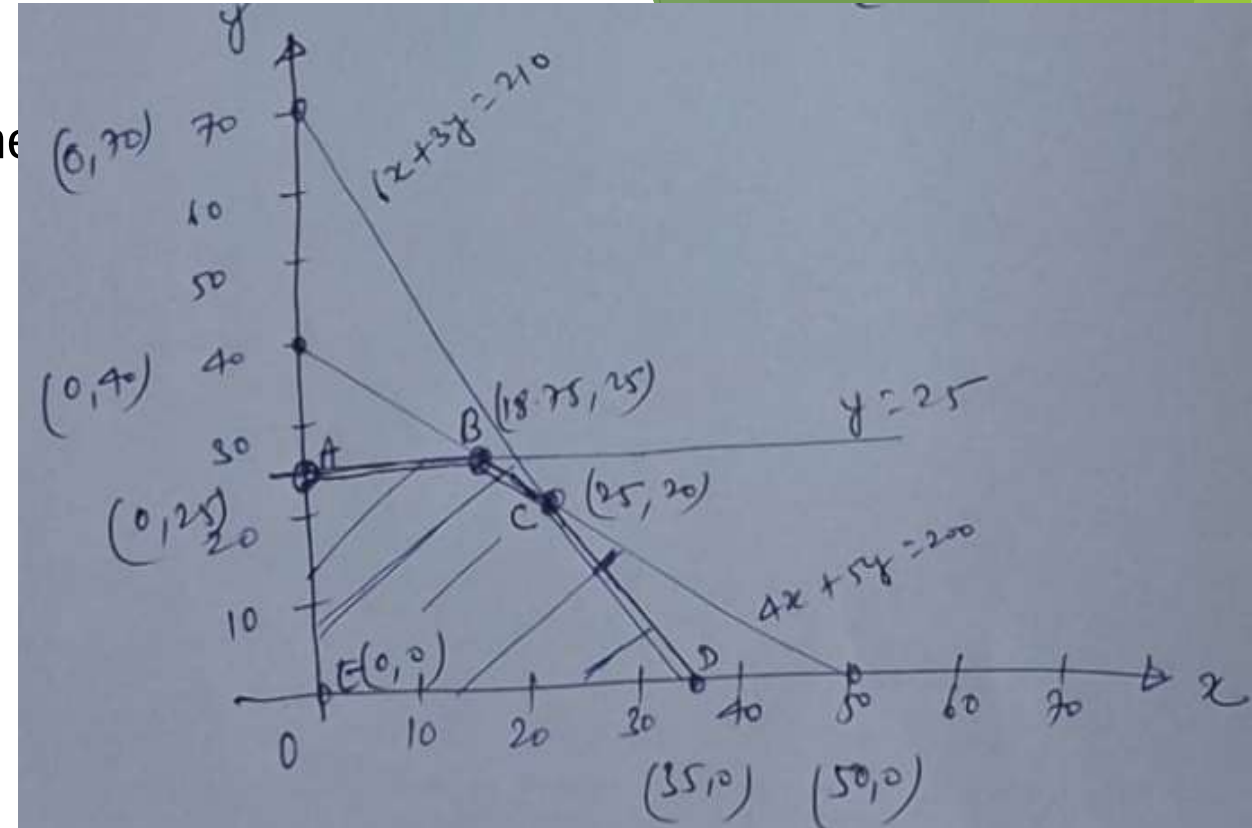
$$6X + 3Y \leq 210$$

$$X, Y \geq 0$$

Again, question is

Whether producing 25 tons of fuel additive and 20 tons of solvent base is still best ??

Decision variables value as well as objective value are changed (Optimal value changed, means producing 25 tons of fuel additive and 20 tons of solvent base is not best)



Sensitivity Analysis

1. How will a change in an *objective function coefficient* affect the optimal solution?

Conclusion

- ✓ If the change in the objective function coefficient is small, the corner/extreme point that provided the optimal solution to the original problem may still provide the optimal solution.
- ✓ However, if the change in the objective function coefficient is large enough, a different corner/extreme point will provide a new optimal solution

Sensitivity Analysis

2. How will a change in a *right-hand-side value* for a constraint affect the optimal solution?

- ✓ What would happen to the optimal solution and total profit contribution if 'RMC inc.' could obtain additional quantities of either of resources?
- ✓ Sensitivity analysis determine how much each added ton of material is worth and how many tons can be added before diminishing return.

Sensitivity Analysis

2. How will a change in a *right-hand-side value* for a constraint affect the optimal solution?

Suppose that in the 'RMC, Inc.' problem an additional 4.5 tons of material 3 becomes available.

$$\text{Max } 40X + 30Y$$

$$0.4X + 0.5Y \leq 20$$

$$0.2Y \leq 5$$

$$0.6X + 0.3Y \leq 25.5$$

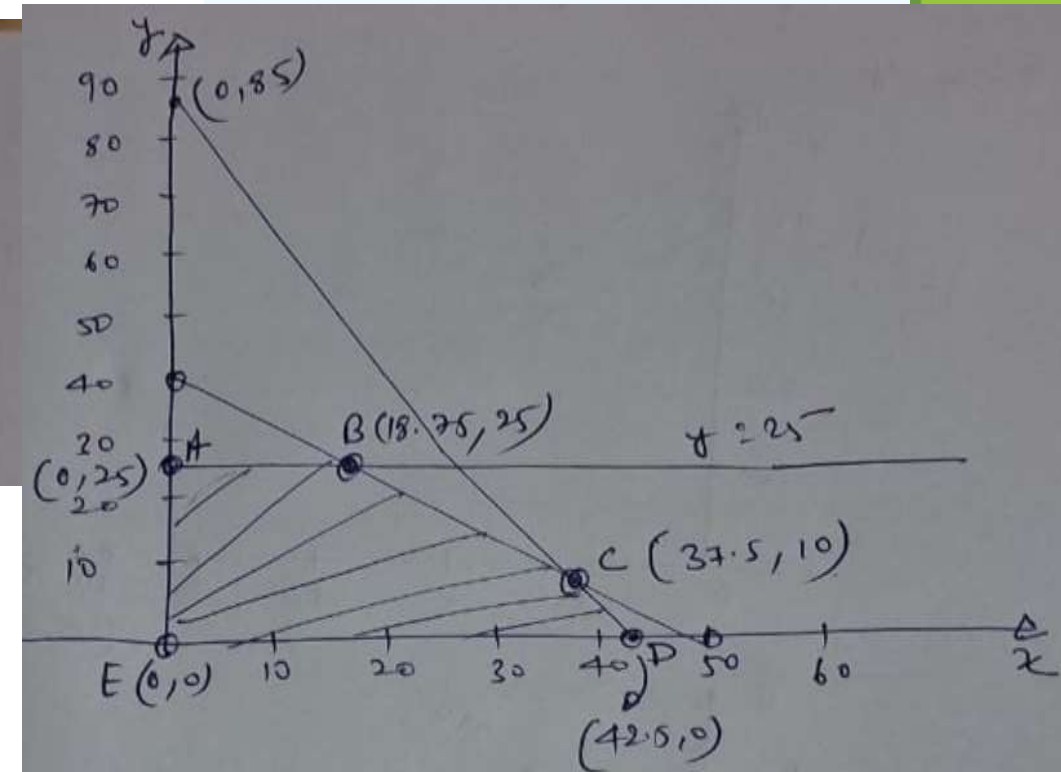
$$X, Y \geq 0$$

$$\begin{aligned} 40x + 30y \\ 4x + 5y = 200 \quad [0, 40] \quad [50, 0] \\ y \leq 25 \\ 6x + 3y = 255 \quad [42.5, 0] \quad [0, 85] \end{aligned}$$

MATERIAL REQUIREMENTS PER TON FOR THE RMC PROBLEM

	Product	
	Fuel Additive	Solvent Base
Material 1	0.4	0.5
Material 2		0.2
Material 3	0.6	0.3

Material	Amount Available for Production
Material 1	20 tons
Material 2	5 tons
Material 3	21 tons



Sensitivity Analysis

- ✓ Whether the feasible region expands or not ? (Yes Expand).
- ✓ As per the graphical solution the corner point $X= 37.5$ tons and $Y= 10$ tons is the new optimal solution.

$$Z = \$1800.$$

- ✓ Thus, the additional 4.5 tons of material-3 in the revised problem provides a new optimal solution and increases the value of the objective by $\$1800 - \$1600 = \$200$.
- ✓ On a per-ton basis, the additional 4.5 tons of material 3 increases the value of the optimal solution at the rate of $\$200/4.5 = \44.44 per ton
- ✓ The **shadow price** is the change in the optimal objective function value per unit increase in the right-hand side of a constraint. Hence, the shadow price for the material 3 constraint is \$44.44 per ton.